SG No. 203 T_{h}^4 $Fd\bar{3}$ [cubic]

* plus set: $+[0,\,0,\,0]$, $+\left[0,\,\frac{1}{2},\,\frac{1}{2}\right]$, $+\left[\frac{1}{2},\,0,\,\frac{1}{2}\right]$, $+\left[\frac{1}{2},\,\frac{1}{2},\,0\right]$

Table 1: Wyckoff site: `8a`, site symmetry: `23.`

No.	position	mapping
1	$\left[\frac{1}{8},\,\frac{1}{8},\,\frac{1}{8}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`
2	$\left[\frac{7}{8},\,\frac{7}{8},\,\frac{7}{8}\right]$	`[13,14,15,16,17,18,19,20,21,22,23,24]`

Table 2: Wyckoff site: `8b`, site symmetry: `23.`

No.	position	mapping
1	$\left[\frac{5}{8},\,\frac{5}{8},\,\frac{5}{8}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`
2	$\left[\frac{3}{8},\,\frac{3}{8},\,\frac{3}{8}\right]$	`[13,14,15,16,17,18,19,20,21,22,23,24]`

Table 3: Wyckoff site: `16c`, site symmetry: `.-3.`

No.	position	mapping
1	$[0,\,0,\,0]$	`[1,5,9,13,17,21]`
2	$\left[\frac{3}{4},\,\frac{3}{4},\,0\right]$	`[2,7,12,14,19,24]`
3	$\left[\frac{3}{4},\,0,\,\frac{3}{4}\right]$	`[3,8,10,15,20,22]`
4	$\left[0,\,\frac{3}{4},\,\frac{3}{4}\right]$	`[4,6,11,16,18,23]`

Table 4: Wyckoff site: `16d`, site symmetry: `.-3.`

No.	position	mapping
1	$\left[\frac{1}{2},\,\frac{1}{2},\,\frac{1}{2}\right]$	`[1,5,9,13,17,21]`
2	$\left[\frac{1}{4},\,\frac{1}{4},\,\frac{1}{2}\right]$	`[2,7,12,14,19,24]`
3	$\left[\frac{1}{4},\,\frac{1}{2},\,\frac{1}{4}\right]$	`[3,8,10,15,20,22]`
4	$\left[\frac{1}{2},\,\frac{1}{4},\,\frac{1}{4}\right]$	`[4,6,11,16,18,23]`

Table 5: Wyckoff site: `32e`, site symmetry: `.3.`

No.	position	mapping
1	$[x,\,x,\,x]$	`[1,5,9]`
2	$\left[\frac{3}{4}-x,\,\frac{3}{4}-x,\,x\right]$	`[2,7,12]`
3	$\left[\frac{3}{4}-x,\,x,\,\frac{3}{4}-x\right]$	`[3,8,10]`
4	$\left[x,\,\frac{3}{4}-x,\,\frac{3}{4}-x\right]$	`[4,6,11]`
5	$[-x,\,-x,\,-x]$	`[13,17,21]`
6	$\left[x+\frac{1}{4},\,x+\frac{1}{4},\,-x\right]$	`[14,19,24]`
7	$\left[x+\frac{1}{4},\,-x,\,x+\frac{1}{4}\right]$	`[15,20,22]`
8	$\left[-x,\,x+\frac{1}{4},\,x+\frac{1}{4}\right]$	`[16,18,23]`

Table 6: Wyckoff site: `48f`, site symmetry: `2..`

No.	position	mapping
1	$\left[x,\ \frac{1}{8},\ \frac{1}{8}\right]$	`[1,4]`
2	$\left[\frac{3}{4}-x,\ \frac{5}{8},\ \frac{1}{8}\right]$	`[2,3]`
3	$\left[\frac{1}{8},\ x,\ \frac{1}{8}\right]$	`[5,8]`
4	$\left[\frac{1}{8},\ \frac{3}{4}-x,\ \frac{5}{8}\right]$	`[6,7]`
5	$\left[\frac{1}{8},\ \frac{1}{8},\ x\right]$	`[9,12]`
6	$\left[\frac{5}{8},\ \frac{1}{8},\ \frac{3}{4}-x\right]$	`[10,11]`
7	$\left[-x,\ \frac{7}{8},\ \frac{7}{8}\right]$	`[13,16]`
8	$\left[x+\frac{1}{4},\ \frac{3}{8},\ \frac{7}{8}\right]$	`[14,15]`
9	$\left[\frac{7}{8},\ -x,\ \frac{7}{8}\right]$	`[17,20]`
10	$\left[\frac{7}{8},\ x+\frac{1}{4},\ \frac{3}{8}\right]$	`[18,19]`
11	$\left[\frac{7}{8},\ \frac{7}{8},\ -x\right]$	`[21,24]`
12	$\left[\frac{3}{8},\ \frac{7}{8},\ x+\frac{1}{4}\right]$	`[22,23]`

Table 7: Wyckoff site: `96g`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$\left[\frac{3}{4}-x,\ \frac{3}{4}-y,\ z\right]$	`[2]`
3	$\left[\frac{3}{4}-x,\ y,\ \frac{3}{4}-z\right]$	`[3]`
4	$\left[x,\ \frac{3}{4}-y,\ \frac{3}{4}-z\right]$	`[4]`
5	$[z,\ x,\ y]$	`[5]`
6	$\left[z,\ \frac{3}{4}-x,\ \frac{3}{4}-y\right]$	`[6]`
7	$\left[\frac{3}{4}-z,\ \frac{3}{4}-x,\ y\right]$	`[7]`
8	$\left[\frac{3}{4}-z,\ x,\ \frac{3}{4}-y\right]$	`[8]`
9	$[y,\ z,\ x]$	`[9]`
10	$\left[\frac{3}{4}-y,\ z,\ \frac{3}{4}-x\right]$	`[10]`
11	$\left[y,\ \frac{3}{4}-z,\ \frac{3}{4}-x\right]$	`[11]`
12	$\left[\frac{3}{4}-y,\ \frac{3}{4}-z,\ x\right]$	`[12]`
13	$[-x,\ -y,\ -z]$	`[13]`
14	$\left[x+\frac{1}{4},\ y+\frac{1}{4},\ -z\right]$	`[14]`
15	$\left[x+\frac{1}{4},\ -y,\ z+\frac{1}{4}\right]$	`[15]`
16	$\left[-x,\ y+\frac{1}{4},\ z+\frac{1}{4}\right]$	`[16]`
17	$[-z,\ -x,\ -y]$	`[17]`
18	$\left[-z,\ x+\frac{1}{4},\ y+\frac{1}{4}\right]$	`[18]`
19	$\left[z+\frac{1}{4},\ x+\frac{1}{4},\ -y\right]$	`[19]`
20	$\left[z+\frac{1}{4},\ -x,\ y+\frac{1}{4}\right]$	`[20]`
21	$[-y,\ -z,\ -x]$	`[21]`
22	$\left[y+\frac{1}{4},\ -z,\ x+\frac{1}{4}\right]$	`[22]`
23	$\left[-y,\ z+\frac{1}{4},\ x+\frac{1}{4}\right]$	`[23]`
24	$\left[y+\frac{1}{4},\ z+\frac{1}{4},\ -x\right]$	`[24]`